Bowen Feng

Ph.D. Candidate, Princeton University · Robot Learning

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Research. I build robots that *reason, learn, and act*: vision-language models whose reasoning is distilled directly into actions, and latent world models that plan and explore under partial observability. Applications span language-steerable navigation, autonomous driving, and multi-agent interaction, validated on real robots and closed-loop simulators.

Education

Princeton University — Princeton, NJ
Ph.D. Candidate, Mechanical and Aerospace Engineering · Advisor: Prof. Felix Heide — *Aug 2022 – present*

Northwestern University — Evanston, IL
M.S. in Robotics — *Sep 2020 – Dec 2021*

Rose-Hulman Institute of Technology — Terre Haute, IN
B.S. in Mechanical Engineering, Minor in Mathematics — *Sep 2016 – May 2020*

Selected Research Publications

[1] **Training Generalist Navigation Policies via Action Self-Distillation.**
Bowen Feng, K. Wang, Y. Yang, A. Joshi, C. Allen-Blanchette, D. Shah, F. Heide.
Conference on Robot Learning (CoRL), 2026.

[2] **EpicWorldModel: Exploration-driven Planning with Latent World Models.**
Bowen Feng, J. Ost, M. Mei, A. Majumdar, F. Heide.
Advances in Neural Information Processing Systems (NeurIPS), 2026.

[3] **VERDI: VLM-Embedded Reasoning for Autonomous Driving.**
Bowen Feng, Z. Mei, J. Ost, F. Ghilotti, B. Li, R. Girgis, A. Majumdar, F. Heide.
arXiv:2505.15925, 2025. (Princeton × Torc Robotics)

[4] **Behavior-Inspired Neural Networks for Relational Inference.**
Y. Yang, **Bowen Feng**, K. Wang, N. E. Leonard, A. B. Dieng, C. Allen-Blanchette.
Int. Conf. on Artificial Intelligence and Statistics (AISTATS), 2025. *PMLR 258.*

[5] **Opinionated Control Barrier Functions for Safe and Efficient Multi-Agent Coordination.**
Bowen Feng*, H. Hu*, M. Santos, J. F. Fisac, N. E. Leonard. *Manuscript.* *equal contribution

Research Experience

Computational Imaging Lab, Princeton University — Princeton, NJ
Graduate Researcher, with Prof. Felix Heide — *2024 – present*

Action Self-Distillation (ASD): language-steerable navigation without human labels — CoRL 2026

- Proposed a self-distillation framework in which a *self-teacher* VLM sees the ground-truth trajectory, infers the goal and writes perception/planning reasoning, and a *student* VLM learns to produce the same reasoning and actions from the raw camera view; the teacher tracks the student via EMA.
- Removes the need for human goal or reasoning annotations; trains on ∼100 h of navigation data, versus 9,500 h for the OmniVLA baseline.
- Deployed **zero-shot on a real FrodoBot rover across 13 tasks**: **84%** indoor success vs. 54% for OmniVLA (94% with a Qwen3.5-VL backbone) and **95%** outdoor; +30 pp over the strongest baselines indoors and on multi-stage instructions.
- Outperforms supervised fine-tuning on frontier-model reasoning labels (80% vs. 36% success on ablation tasks).

EpicWorldModel: exploration-driven planning with stochastic latent world models — NeurIPS 2026

- Replaced deterministic JEPA world models with stochastic JEPAs trained by flow matching, predicting multiple plausible futures when goal-relevant content is occluded.
- Used flow predictive variance as an exploration signal inside CEM planning, trading off goal-reaching against exploring uncertain regions; up to **22%** higher success than LeWorldModel.

VERDI: distilling VLM reasoning into modular end-to-end driving Princeton × Torc Robotics

- Aligned the perception, prediction, and planning modules of an end-to-end driving stack with VLM text features at training time, embedding commonsense reasoning with *no* VLM cost at inference.
- Up to **11%** better ℓ_2 and driving performance than e2e methods without embedded reasoning on nuScenes, Bench2Drive, and closed-loop HugSim, at real-time speed.

Continual skill optimization for autonomous driving 2024

- Developed lifelong policy optimization with Elastic Weight Consolidation so new driving skills (e.g., pedestrian interaction) are learned without forgetting existing ones.

Leonard Lab, Princeton University Princeton, NJ
Graduate Researcher, with Prof. Naomi Ehrich Leonard *Aug 2022 – present*

- Coupled graph neural networks with nonlinear opinion dynamics to infer interpretable, overlapping relations among interacting agents, improving long-horizon trajectory prediction *(AISTATS 2025)*.
- Combined nonlinear opinion dynamics with control barrier functions for safe, efficient two-agent coordination at traffic intersections.

Interactive & Emergent Autonomy Lab, Northwestern University Evanston, IL
Graduate Researcher, with Prof. Todd Murphey and Dr. Matthew Elwin *Mar 2021 – Mar 2022*

- Used a robot arm to bend flexible wire into target shapes as tools, via optimal control and active learning over the wire's equilibrium-shape manifold.

Selected Projects

- **Deep Q-learning for urban driving (CARLA):** end-to-end collision-free driving learned without hand-coded traffic or lane-following rules.
- **Neural Circuit Policies on a Clearpath Jackal:** CNN + NCP navigation (obstacle avoidance, line following) with far fewer control neurons and less data.
- **Mobile manipulation, KUKA youBot (simulation):** trajectory planning, odometry, and feedforward + PID control for pick-and-place.
- **Automated 3D modeling:** Sawyer arm and TurtleBot3 cooperatively scan objects with a RealSense camera for CAD-free reconstruction.

Skills

Learning: VLM fine-tuning & distillation (Qwen-VL), reinforcement learning, world models, imitation learning, PyTorch, Python
Robotics: real-robot deployment (FrodoBot rover, Clearpath Jackal, Sawyer, TurtleBot3), ROS, optimal control, control barrier functions, motion planning
Simulation & data: CARLA, HugSim, Bench2Drive, nuScenes